

AUGUSTIN RAJU

Cologne, Germany

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Professional Summary

Mechatronics engineer with expertise in robotics, AI and autonomous systems. Brings a research mindset to engineering challenges, with a track record of delivering solutions that work beyond simulation and into the real world.

Experience

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|-------------------|--|
| 09/2025 – Present | Research Associate - TH Cologne, Cologne <ul style="list-style-type: none">• Working under Smart Automation team of Innofaktur project to support regional companies• Developed robotics applications using ROS2 and Python• Built a VR based robot teaching application using Unity and C# |
| 04/2024 – 04/2025 | AI Intern - Volkswagen AG, Wolfsburg <ul style="list-style-type: none">• Developed a Physics Informed Neural Network virtual sensor for wheel parameter prediction• The same model could be used as a simulator to mimic actual vehicle behaviour• Developed a trajectory tracking controller using RL which outperformed existing controller in simulation |
| 10/2020 – 08/2021 | Project Associate - IIST, Kerala, India <ul style="list-style-type: none">• Developed a rover simulation system using a rigid body dynamics algorithm• Simulated wheel- ground interactions using Terramechanics models |

Education

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| 2025 | M.Sc. Mechatronics - University of Siegen, Siegen
Grade: 1.5 |
| 2020 | B.Tech Mechanical Engineering - APJKTU, Kerala, India
Grade: 1.6 |

Skills

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|------------------------------|---------------------------------------|
| Languages | English (C1), German (B1) |
| Programming Languages | Python, Arduino, MATLAB, Shell, Bash |
| Software Tools | Docker, Git, CAD, Unity |
| Frameworks | ROS2, Nav2, MicroROS, PyTorch, RLLib |
| Hardware | Jetson, Teensy, Raspberry Pi, Arduino |

Achievements

RoboCup Smart Manufacturing League German Open 2026 Winner

My team developed a pick and place mobile robot to work in a model future factory scenario

Shell Eco Marathon Asia 2020 – Circular Economy Award

My team developed a prototype vehicle for an energy efficiency competition, receiving an award for sustainable vehicle design